Learning notes: visualising molecules

photocatalyst-dft, week 1 — Oct 2026 — tools: RDKit, py3Dmol, PySCF (+ Claude / Claude Science for coding)

Why I'm writing this. I can now make nice pictures of molecules, but the point of these notes is to write down *why* each picture exists and what it fooled me with. One sentence to remember:

Remember: every picture answers one question, and should come with a number. Otherwise I'm just looking at what the isovalue / viewing angle / colours decided to show me.

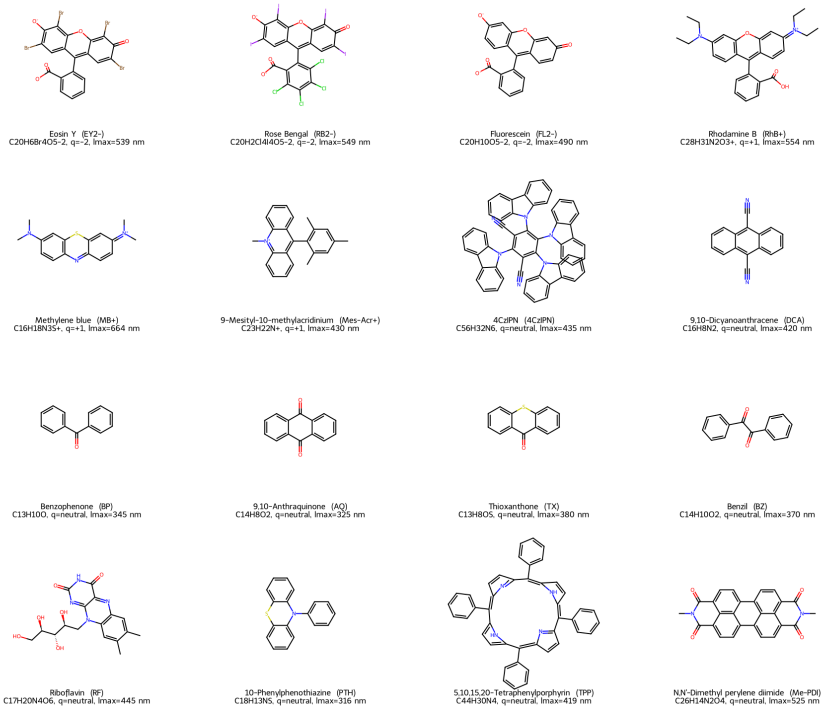
My mental map so far:

- **2D** (RDKit, from SMILES) → "is this the right molecule and charge?"
- **3D** (py3Dmol, from xyz) → "is the shape sensible: flat, twisted, crowded?"
- **overlay** (two xyz files) → "how much did DFT move the force-field structure?"
- **orbitals** (PySCF → cube file → viewer) → "where does the electron leave from / go to?"

1. 2D pictures: checking what I typed

Q: Why bother drawing a 2D picture if I'm going to do 3D anyway?

Because it's the cheapest check there is, and it catches the most expensive mistakes. A SMILES with a missing bromine or the wrong charge still runs. DFT will happily spend hours on a molecule that doesn't exist.



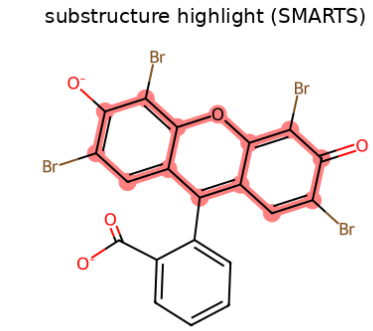
My 16 photocatalysts. The first thing I check now is the little charge signs.

Aha

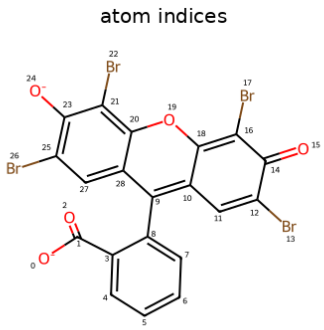
The "same" dye isn't one molecule. Eosin Y in the bottle is neutral, but in water at working pH it's the **dianion** EY^{2-}. The colour and redox behaviour belong to the dianion, so that's what I must compute. Same story for fluorescein and rose bengal (dianions), and methylene blue, rhodamine B and Mes-Acr^{+} (cations). The 2D drawing makes these charges visible at a glance.

Two RDKit things I didn't know before:

- **SMARTS = a search query for a substructure.** I highlighted the xanthene core and suddenly saw that eosin Y, fluorescein, rose bengal and rhodamine B are all the same chromophore with different things attached. The ~ in the pattern means "any bond". It lets the query match whatever bond orders the SMILES uses (here the xanthene is drawn with quinoid bonds, not as plain aromatic rings).
- **Atom indices.** When something says "atom 12" I can draw the numbers. The xyz file uses the same order, which is how I picked four atoms for a dihedral angle later.



eosin Y dianion - xanthene chromophore highlighted



eosin Y - RDKit atom indices

Note to self: export SVG not PNG for Overleaf, since it stays sharp.

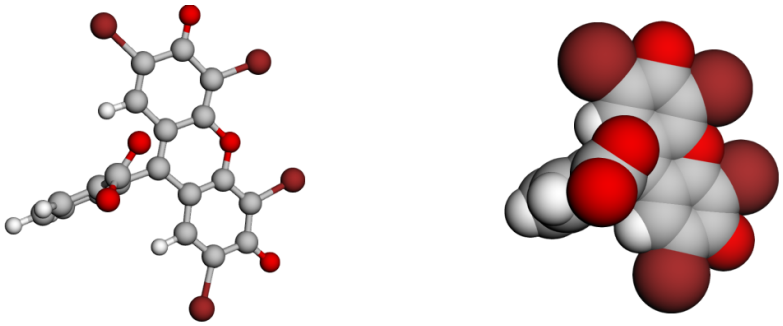
2. 3D pictures: the shape matters more than I expected

How the coordinates are made: SMILES → add H → ETKDGv3 (random distance geometry + torsion preferences from crystal structures) → MMFF94s force-field clean-up. The final version tries 8 conformers (30 for floppy molecules) and keeps the lowest.

Q: Why several conformers? Isn't one structure enough?

For rigid dyes, one is fine. For floppy ones (riboflavin's tail, rhodamine's ethyl groups) a single random embedding can get stuck in a bad shape. One conformer = one sample, not "the" structure.

Styles: **ball-and-stick** for checking bonds and angles; **space-filling** for real size and crowding; **stick/line** when the molecule is big.

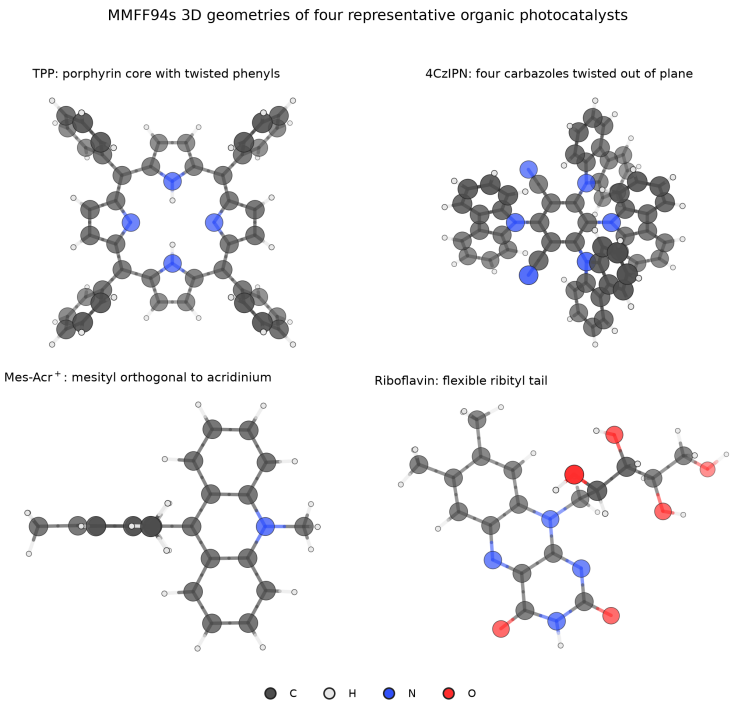


Eosin Y, same coordinates. Left: the benzoate ring is edge-on, i.e. twisted. Right: the four bromines take up a lot of the volume, which I didn't appreciate from 2D.

Aha

Almost every interesting 3D feature in my set is a **twist**, and twisting controls **conjugation**. Two π systems at 90° can't share electrons, so they act as separate pieces.

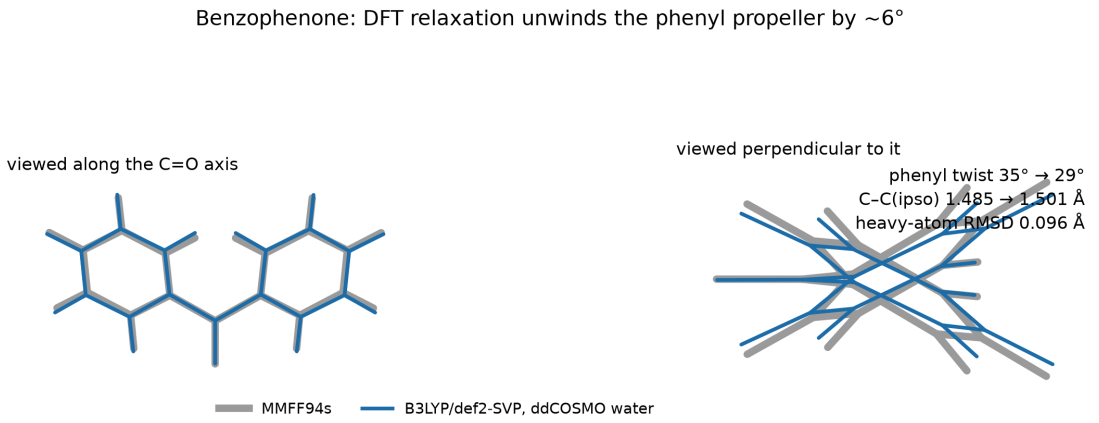
- eosin Y: benzoate ≈ perpendicular to the xanthene (I measured **98.6°**) → the benzoate hardly affects the colour.
- TPP: phenyls twisted out of the porphyrin plane → the macrocycle is the chromophore.
- 4CzIPN: carbazoles twisted → donor and acceptor decoupled → TADF / photoredox.
- Mes-Acr$^+$: ortho-methyls lock the mesityl orthogonal to the acridinium → donor and acceptor in one molecule but kept apart (comes back in Section 4!).



Remember: measure, don't eyeball. "Looks perpendicular" → dihedral angle. "Looks the same" → RMSD.

3. Overlays: did DFT actually change anything?

Idea: draw the MMFF start and the DFT result on top of each other.



Aha

The left view (along C=O) says "nothing changed". The right view shows the phenyl propeller unwound from 35° to 29°. **The viewing angle can completely hide a change.** Always look from two directions, and trust the RMSD more than my eyes.

What the numbers told me:

	heavy-atom RMSD (Å)	opt. cycles	what it means
anthraquinone	0.019	15	rigid → MMFF basically right
thioxanthone	0.030	10	rigid → MMFF basically right
benzophenone	0.096	11	2 rotatable rings → 5× worse

Mistake I caught

We used these three results to say "MMFF is good enough" and then skipped DFT optimisation for the other 13 molecules. But the three tested were mostly *rigid*, and the one floppy-ish case was already 5× worse. The force field is weakest exactly at the torsions between rings, which is where conjugation is decided. So I can't extend that conclusion to rhodamine B, riboflavin or the charged dyes without optimising them. *To do.*

4. Orbitals: the bit I found hardest

How the picture is made: PySCF evaluates the orbital on a 3D grid → writes a *cube file* (plain text; py3Dmol, VESTA, VMD, Avogadro all read it) → draw two surfaces, +iso and −iso, in two colours.

Q: What do the two colours mean?

Just the sign of the wavefunction. They're **arbitrary**: ψ and $-\psi$ are the same orbital. What matters is where the colour *changes*, which is a **node** (zero amplitude).

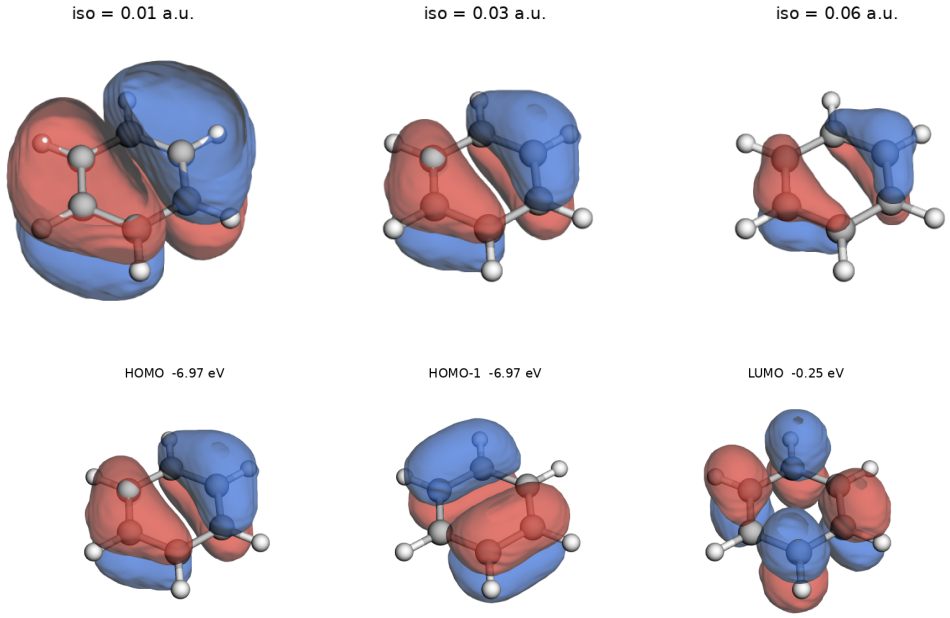
Q: How big should the blobs be?

That's the isovalue, and it's my choice. I drew the same orbital at three values:

Aha

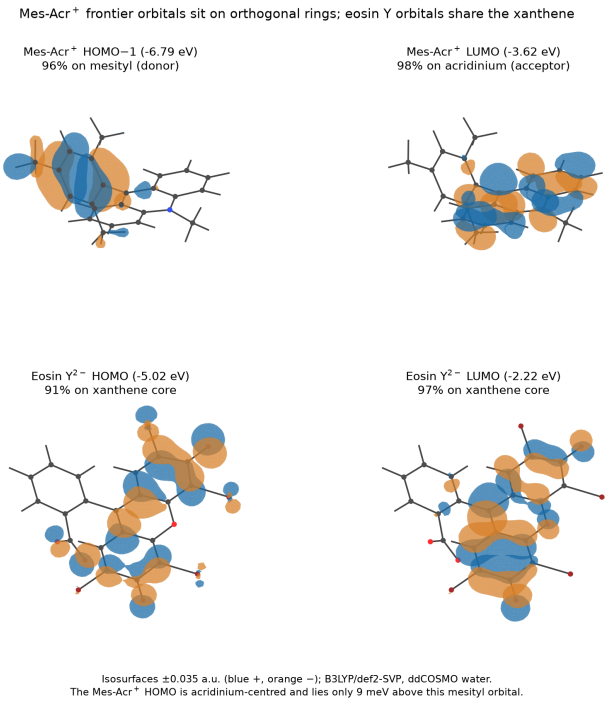
The lobes change size but **the nodes don't move**. Nodes are physics; lobe size is a display setting. So: same isovalue for every orbital I compare, and never argue from how "big" a lobe looks.

Benzene as a sanity check (things I can check against the textbook):



- HOMO and HOMO−1 have the *same* energy (−6.97 eV): a degenerate pair (e_{1g}). The program's choice of which way the nodal plane points is arbitrary. **Don't interpret one half of a degenerate pair alone.**
- LUMO has two nodal planes vs one for the HOMO. More nodes, higher energy: the particle-on-a-ring picture from first year, finally seen directly.

Then the real catalysts:



Since a picture depends on the isovalue, we also computed *how much* of each orbital sits on each fragment (Mulliken populations):

	orbital	E (eV)	where
eosin Y^{2-}	HOMO	−5.02	91% xanthene
	LUMO	−2.22	97% xanthene
Mes-Acr^+	HOMO−2	−6.80	>99% mesityl
	HOMO−1	−6.79	96% mesityl
	HOMO	−6.79	99% acridinium
	LUMO	−3.62	98% acridinium

Aha

Eosin Y: electron starts and ends on the xanthene → *local* $\pi \to \pi^*$.
Mes-Acr^+: electron can go from the mesityl (donor) to the acridinium (acceptor), two rings that the 3D picture showed are at 90° → *charge transfer*. The 3D shape and the orbital picture are telling the same story. That's the moment it clicked for me.

Mistake I caught

I was about to write "the Mes-Acr^+ HOMO is on the acridinium". But the top three occupied orbitals are within **13 meV** of each other, far below what B3LYP can resolve. Which one gets the label "HOMO" is basically random at this level. Look at the set, not the label.

Caveats I need to keep in mind:

- Mulliken % depends on the basis set, so use it for "91% vs 10%", not "91% vs 88%".
- B3LYP is bad at charge-transfer excitations (Mes-Acr^+, 4CzIPN) → for TDDFT use CAM-B3LYP or ωB97X-D.

5. The energy-level diagram: the picture that fooled us

Plotting HOMO/LUMO for all 16 side by side is nice for trends: the HOMO–LUMO gap follows the lowest absorption band ($r = 0.90$, MAE 0.45 eV). Biggest outlier: benzophenone (+1.37 eV), because its weak first band is n→ π^*, not $\pi \to \pi^*$.

Mistake I caught

Our diagram has the H^+/H_2 and O_2/H_2O lines drawn across the orbital energies. That *looks* like it tells me which dyes can split water. It doesn't. From benzene I know −HOMO is ∼2 eV off the real ionisation energy with B3LYP, and the HOMO–LUMO gap isn't the optical gap either. A good correlation means the *trend* is right, not that the absolute energies are. → Remove the lines or label them "qualitative". Redox needs ΔSCF in solvent; absorption needs TDDFT.

6. Practical things I learned the hard way

- **py3Dmol can't save a PNG on its own**: it needs a browser. Workarounds: a headless browser (Playwright), or redraw with matplotlib marching cubes (what Claude Science did). The matplotlib ones only roughly handle which surface is in front, so they're schematic. For real figures: VESTA or VMD.
- **pyscf-dispersion didn't load on my Mac** → no D3 correction in these runs. Must write that next to every result.
- **Cube files are 3–5 MB each** → `*.cube` in `.gitignore`; they regenerate in seconds.

- **AI-written code**: fine, but every script got one test against a known answer (benzene, water) before I trusted it on a dye.

7. Test myself

Test myself

1. Why is eosin Y computed as EY^{2-} and not as the neutral molecule?
2. What does ~ mean in a SMARTS pattern, and why use it here?
3. Why does the benzoate ring in eosin Y barely change its colour?
4. Why can one viewing direction hide the benzophenone change completely?
5. If I swap blue and red on an orbital plot, what changes?
6. The lobes shrink when I raise the isovalue. What stays the same?
7. Why shouldn't I interpret only the HOMO of benzene?
8. What links Mes-Acr^+'s 3D shape to its orbital picture?
9. Why is "the HOMO of Mes-Acr^+" not a meaningful label here?
10. Why can't I read water-splitting ability off the energy-level diagram?

Still unsure / ask Prof. Hu

- Is Mulliken OK for fragment %, or should I use something less basis-dependent (Hirshfeld, Löwdin)?
- For the floppy dyes, is a quick xTB conformer search good enough before DFT?
- Which isovalue does the group normally use in figures?

Next time I'll...

1. draw 2D and check the charge before anything else;
2. measure dihedrals / RMSD instead of saying "looks fine";
3. check degenerate or near-degenerate orbitals before naming "the HOMO";
4. use one isovalue throughout and say which;
5. label every figure with functional / basis / solvent (here: B3LYP/def2-SVP, ddCOSMO water, no D3).